Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI Agents · Human-in-the-Loop Workflows · Evaluation Infrastructure · Context Management · Prompting Systems · Full-Stack · Python · AWS

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SUMMARY

Software engineer specializing in **AI agent systems**, human-in-the-loop workflows, evaluation infrastructure, and full-stack product surfaces. Currently at Intuit building Anthropic Claude integrations into production financial services: **autonomous AI workflows** with tool calling, prompting systems, structured outputs, and guardrails; **evaluation harnesses** with LLM-as-judge scoring, golden datasets, and CI regression gates; monitoring via structured logging and production quality signals. Built **Leasel** - a production agentic platform (LangGraph, 8-agent orchestration) with durable context management, multi-step conditional routing, **HITL guardrails** for regulated decisions, and three product surfaces sharing one AI and retrieval layer (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source **evaluation and testing infrastructure** platform with offline golden-dataset evals, LLM-as-judge scoring, online quality metrics, CI regression gating, and full per-run tracing. At Entrata (7+ years, 100K+ users), owned mission-critical async distributed systems and data pipelines in regulated multi-tenant SaaS. Strong full-stack: Python/FastAPI backend + React/TypeScript frontend. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

AI Teammates · Autonomous Agents · Agentic Workflows | Human-in-the-Loop Workflows · HITL · Guardrails

Evaluation & Testing Infrastructure · LLM-as-Judge · CI Regression | Context Management Over Time · Durable State · Multi-Step Workflows

Prompting Systems · Structured Outputs · Tool Calling | Agent Capabilities · Multi-Agent Orchestration · LangGraph · LangChain

Product Surfaces · Interaction Patterns · Cross-Functional Collaboration | Full-Stack: Python · FastAPI · React · TypeScript · Node.js

Observability · Tracing · Structured Logging · OpenTelemetry | AWS (Lambda · Bedrock · SQS · API Gateway) · Docker · Kubernetes

Distributed Systems · Async Processing · Backend Engineering | End-to-End Ownership · Rapid Experimentation · Production Quality

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **AI agent workflows in production:** Build LangChain-based autonomous AI workflows integrating Anthropic Claude into production financial services - prompting systems with tool calling and retrieval-grounded context management, structured-output schemas, third-party platform integrations; guardrails with input validation, sensitive-data handling, and fallbacks; HITL escalation for sensitive decisions.
- **Evaluation and testing infrastructure:** Build LLM-as-judge evaluation harnesses with golden datasets and CI regression gates; monitor production AI behavior via structured logging and CloudWatch quality signals; perform qualitative error analysis on failures and convert them into eval coverage that prevents regression. Document interaction patterns and playbooks for team reuse.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript product surfaces; deploy via Docker + GitHub Actions CI/CD to AWS with full ownership from data layer through UI.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned mission-critical data pipelines and async distributed systems end-to-end - HUD compliance ETL (180K records, zero violations across 6 years), RabbitMQ event-driven processing with idempotency guarantees, PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - AI Agent Platform: HITL · Context Management · Multi-Step Workflows · Product Surfaces

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI agent platform: **8-agent LangGraph orchestration** with tool/function calling, **durable context management over time**, conditional routing, and multi-step autonomous workflows; **HITL guardrails** routing regulated decisions to humans for trust and safety; **hybrid search RAG** (OCR ingestion, TF-IDF + semantic reranking, citation grounding); **CI eval harness** (LLM-as-judge, golden datasets: 93.6% precision, 87.9% F1 across 1,240 test cases). Three distinct product surfaces share one AI and retrieval layer - built for re-use and cross-functional extension.

AgentProbe - LLM Evaluation & Testing Infrastructure

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source **evaluation and testing infrastructure**: offline golden-dataset evals, LLM-as-judge scoring, online quality metrics, **CI regression harness** gating releases on eval thresholds, failure handling with structured error classification; cost-aware routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing and structured logging. Used as a production release gate before every AI deploy.

AI on FHIR — Conversational AI & Third-Party Platform Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. Full-stack: Python/Flask backend + React frontend; OAuth-secured API integration, HIPAA-style RBAC, Docker-containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017